5.1 Solving Systems of Linear Equations by Graphing

Name _____ Period ____ Date _____

Algebra I 5.1 · Student sheet

I can…

- check whether an ordered pair is a solution of a system.
- solve a system of linear equations by graphing.
- write and solve a system of equations for a real-life problem.

Vocabulary · fill in the missing word

System of linear equations — two or more linear equations with the same _____

Solution of a system — an ordered pair that makes every _____ true; where the graphs intersect

Part 1 · Two plans

Plan A: $10 per visit. Plan B: $30 plus $5 per visit.

Visits	0	2	4	6	8
Plan A					
Plan B					

Graph both plans and mark where they meet.

When do they cost the same? _____ When is Plan A better? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Solution of a system: A solution of a system makes _____ equation true.

Check: Check the point in _____ equations.

Example 1

Is (2, −1) a solution of the system?

$2x + y = 3$

$x - 3y = 6$

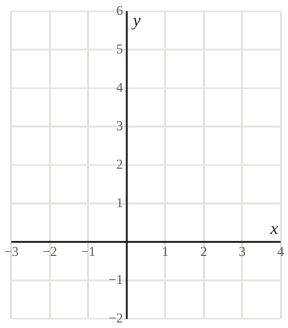
On a graph: On a graph, the solution is where the lines _____.

Example 2

Solve by graphing.

$y = x + 1$

$y = -2x + 4$



Example 3

Plan A: C = 10v. Plan B: C = 5v + 30. For how many visits do the plans cost the same?

Part 3 · Partner practice

Partner 1

Is (3, 1) a solution of $x + y = 4$ and $2x - y = 5$?

Partner 2

Solve by graphing: $y = 2x - 3$ and $y = -x + 3$.